

# Generating the Cloud Motion Winds Field from Satellite Cloud Imagery Using Deep Learning Approach

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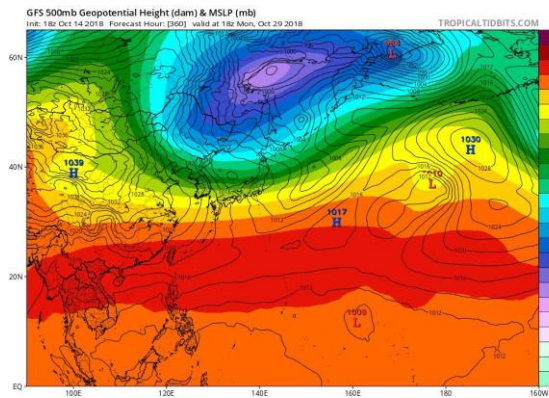
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# Applications of Cloud Motion Winds

**Cloud Motion Winds (CMW):** CMW algorithm estimates the average motion vectors of the atmosphere in a certain area based on the movement of the same cloud mass in two consecutive infrared satellite cloud imageries.



a. weather analysis / numerical weather prediction



b. ultraviolet index



c. solar energy prediction guidance



# Motivation & Challenges

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## Motivation:

- similar to the task of dense optical flow estimation.
- unsupervised deep learning approach.
- automatically feature representation.
- efficient computing speed.

## Challenges:

- the brightness consistency is not guaranteed
- more robust performance is required
- big data training manner.

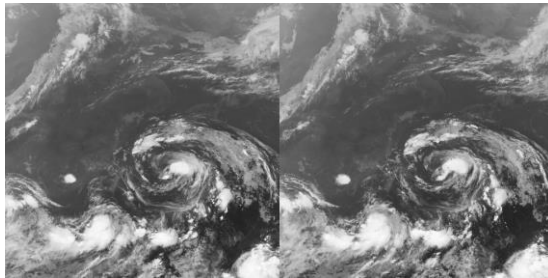


# Cloud Motion Winds Dataset



**CMWD:** The first large-scale **C**loud **M**otion **W**inds **D**ataset for deep learning research

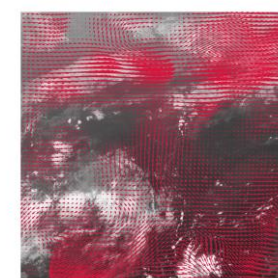
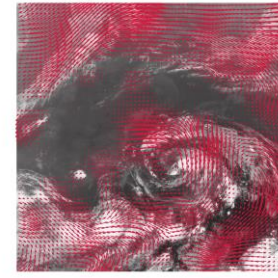
- 6388 training pairs and 715 testing pairs.
- taken over a vast of the Pacific Northwest.
- high resolution and rich cloud cover.



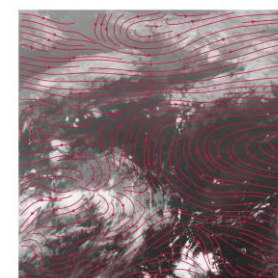
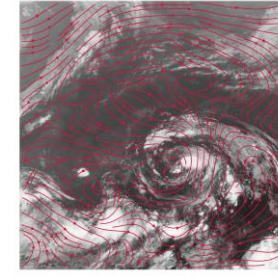
training pairs in CMWD dataset



visualized in HSV  
color space



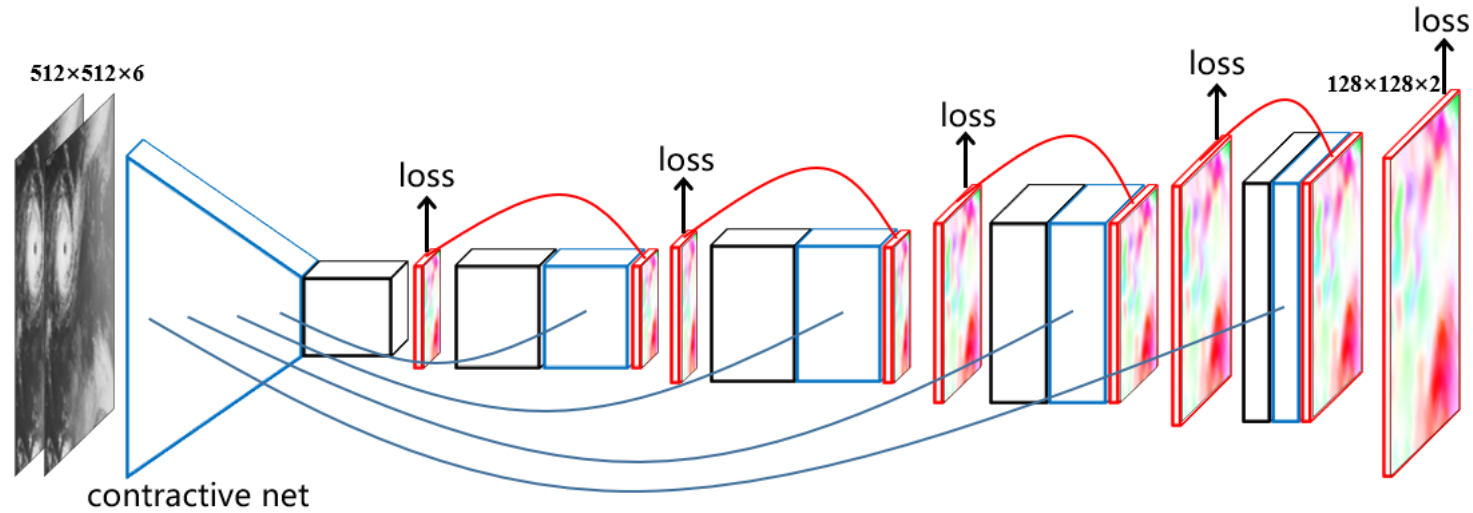
flux streamline  
diagram



dense vector  
arrow diagram



# Network Structure

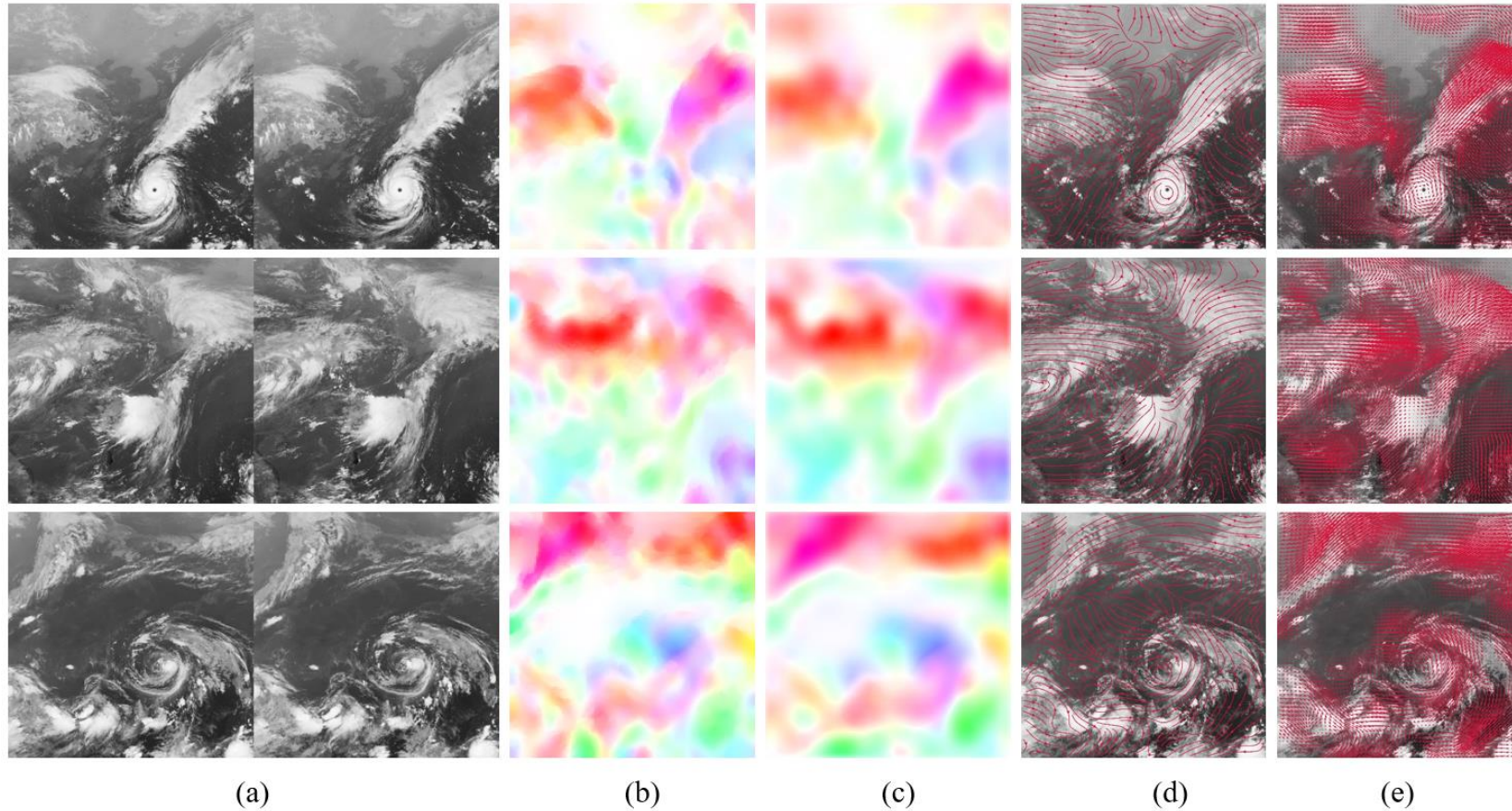


The network architecture of **FlowNetS** model. Two consecutive geostationary meteorological satellite cloud imageries are taken as input, and a cloud motion winds vector field is generated using a multi-stage refinement process. Feature maps from the contractive part, as well as intermediate motion vector field predictions, are used in the “up convolutional” part.





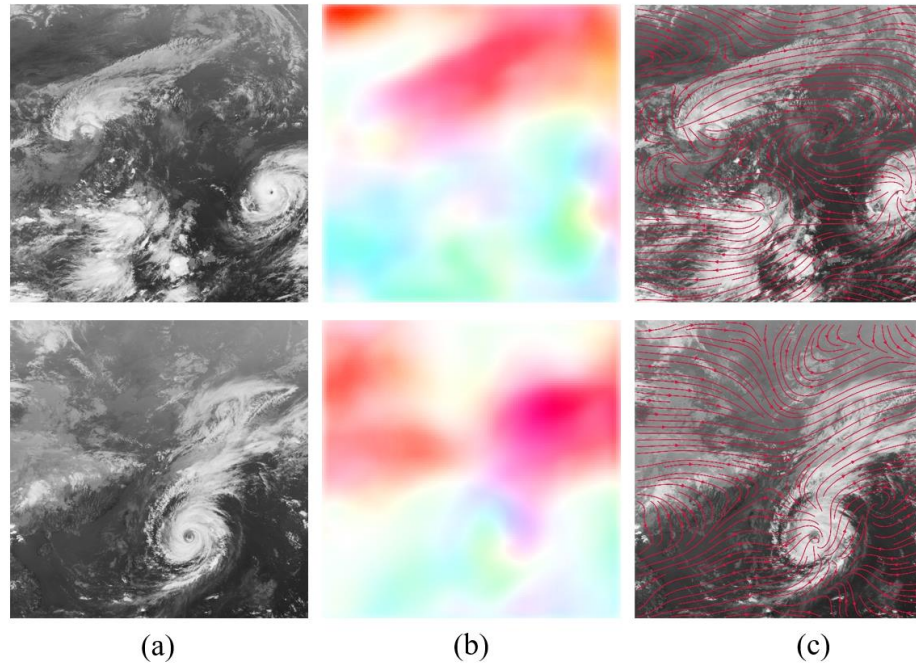
# Experiments on Paired Cloud Imagery



Three experimental results of test samples on CMWD. **(a)** is the input two infrared cloud imageries. **(b)** and **(c)** are the ground truth and predicted visual flow field maps respectively. **(d)** is the generated flux streamline diagram, and **(e)** is the dense vector arrow diagram.



# Experiments on Single Cloud Imagery

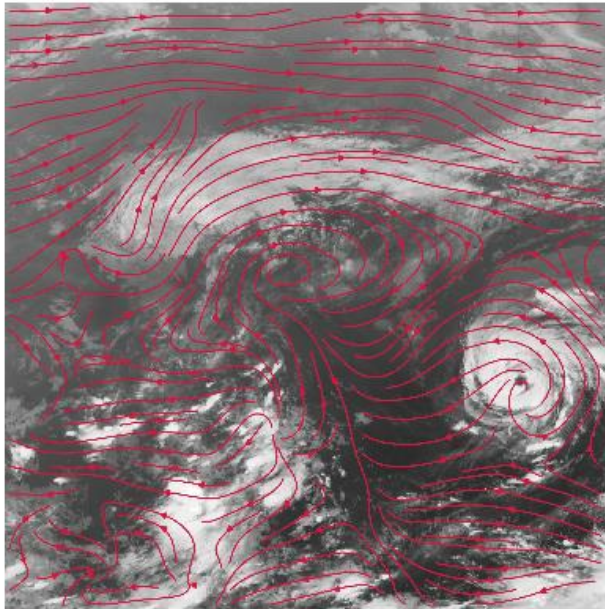


Two experimental results on single cloud imagery. **(a)** is the input single cloud imagery. **(b)** is the predicted visual flow field map and **(c)** is the generated flux streamline diagram.

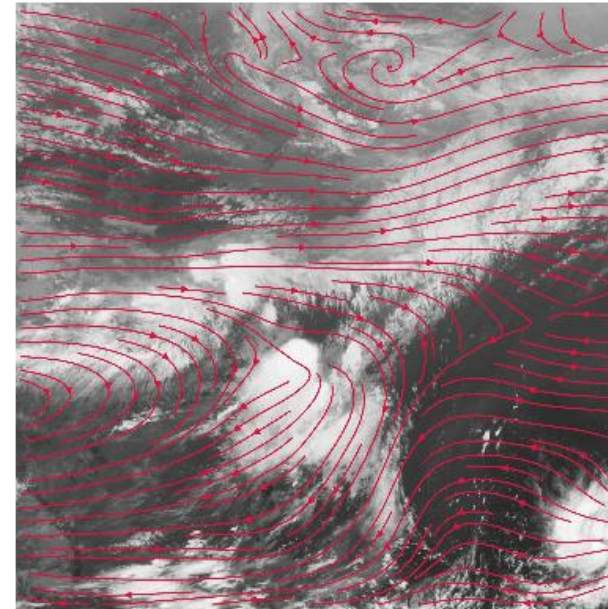




# Experimental Visualization



two-frame predicted flux streamline diagram (gif)



single frame Predicted flux streamline diagram (gif)





## Conclusions

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- Deep convolutional neural network is used to estimate the cloud motion winds field.
- A large-scale cloud motion winds dataset is proposed for deep learning research.
- Try to use single infrared cloud imagery to estimate the cloud motion winds field.



# THANKS



- source code
- dataset
- pre-trained model



- project page